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Tyler Layton

Education

Rensselaer Polytechnic Institute

Class of 2026

Troy, NY | M.S/B.S. Computer Science | B.S. Computer and Systems Engineering | GPA: 3.68

Relevant Coursework: Embedded Controls, Intro to Algorithms, Circuits, Robotics, Eng Probability, Signals & Systems, Principles of Software, Intro to AI, Intro to Electronics, Comp Architecture/Networks & Operating Systems, Design and Analysis of Algorithms, Programming Languages

Experience

Graduate Project

May 2025 – May 2026

Rensselaer Polytechnic Institute

Troy, NY

- Currently designing a web application for 2D network visualization, enabling users to create, customize, and analyze large graphs with search algorithms such as DFS, BFS, and Dijkstra's.

Frontend Engineer

January 2025 – May 2025

Design Lab – Multidisciplinary Capstone Project

Troy, NY

- Worked with a multidisciplinary team of five other RPI students to continue a project sponsored by Lockheed Martin and in association with Georgia Tech and Purdue University.
- Improved user efficiency by 20% by enhancing the UI using TypeScript/HTML/CSS and integrating new technologies into the application to connect the frontend to the Django backend.
- Communicated project progress to the Lockheed Martin sponsors through structured cross-disciplinary updates and weekly sprint reviews.

Software Engineering Intern

Summer 2023 & Summer 2024

L3Harris

Clifton, NJ

- Worked in the Space and Airborne Systems department across two summers, designing and improving internal applications to streamline engineering workflows and save company time and money.
- Led a team of interns to design a full-stack desktop application from scratch in Python with Tkinter for creating and managing Jenkins files, reducing file authoring time by 80%.
- Improved the user experience of an internal Java-based "Message Analyzer" tool for interpreting signals between endpoints, increasing scripting readability and efficiency by 30%.
- Strengthened skills in full-cycle application development, Python, Java, UI/UX design, teamwork, and collaboration with engineers and other interns.

Frontend Engineer/Project Lead Developer

Fall 2022 & Fall 2023

Rensselaer Center for Open Source (RCOS)

Troy, NY

- Led and contributed to open-source projects supporting student learning, including an educational analog circuit simulator and a Python-based platformer game using Pygame.
- Developed frontend UI for a circuit simulation tool to improve student understanding of analog design, applying academic knowledge to create accurate and intuitive features.
- Directed a small team to design and implement user-friendly gameplay mechanics, such as platform jumping, while strengthening leadership and collaborative development skills.
- Enhanced abilities in Python, front-end engineering, teamwork, and project leadership.

Technical Skills

- **Languages/Frameworks:** Python, Java, C++, C, C#, JavaScript, TypeScript, RStudio, Git, MIPS, Haskell, Django
- **Other Skills:** Microsoft Excel, PowerPoint, VS Code, GitHub, MATLAB, NX, LTspice, Eclipse, Windows Form Applications, Unity, Photoshop, Jenkins, Node.js, React, HTML/CSS